

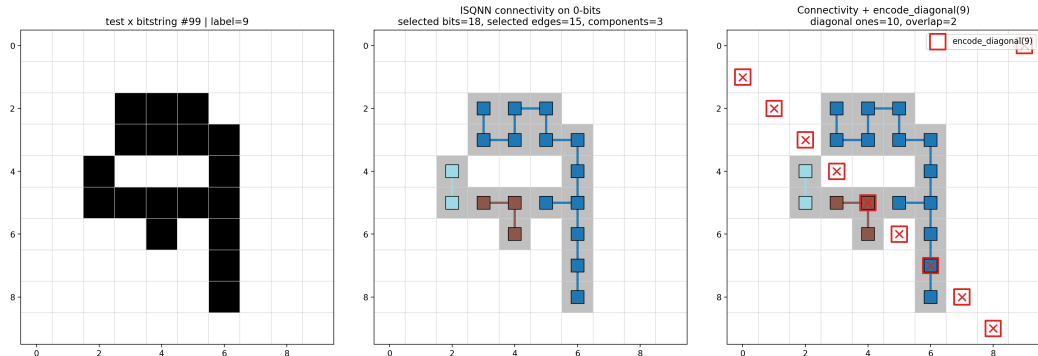
Why Adding R_x Gates Cannot Rescue MNIST Digit Recognition

A Diagnostic Summary

Quantum Machine Learning

April 29, 2026

Original Problem: Insufficient Generative Ability



- Original encoding: $x_j = 0 \rightarrow |+\rangle$, $x_j = 1 \rightarrow |0\rangle$; $|0\rangle$ is a CZ eigenstate and acts as a barrier.
- The available ISQNN connectivity is fragmented into local components, while the diagonal label target is global and sparse.
- The original circuit does not possess the ability to generate global patterns.

Proposed Fix: $R_x(\phi)$ Gates

$$R_x(\phi)|0\rangle = \cos \frac{\phi}{2}|0\rangle - i \sin \frac{\phi}{2}|1\rangle \equiv |u\rangle$$

- Insert $R_x(\phi)$ after encoding, before $R_z(\theta)$
- $|u\rangle$ has $|1\rangle$ component $\Rightarrow$ CZ can now entangle with neighbors $\Rightarrow$ barrier broken

What is guaranteed, and what is not

The circuit's expressivity is certainly improved: $x = 1$ is no longer a strict CZ barrier. However, whether this improvement is significant enough for MNIST, and whether it introduces new estimation or inference problems, must be verified.

Estimation When All Neighbors Are $x = 1$

Consider target qubit j with $x_j = 0$, connected to d CZ neighbors, all with $x = 1$ ($|0\rangle \xrightarrow{R_x} |u\rangle$).

After encoding, $R_x(\pi/4)$, $R_z(\theta_j)$, and CZ, the reduced density matrix of qubit j **in the Z basis** $\{|0\rangle, |1\rangle\}$ (before final H gate):

$$\rho_j = \frac{1}{2} \begin{pmatrix} 1 & (\cos \phi)^d e^{-i\theta_j} \\ (\cos \phi)^d e^{i\theta_j} & 1 \end{pmatrix}$$

Converting to the X basis via $\langle \pm | \rho_j | \pm \rangle$:

$$P(y_j = 1) = \frac{1}{2} [1 - (\cos \phi)^d \cos \theta_j]$$

Inverting:

$$\hat{\theta}_j = \arccos\left(\frac{1 - 2\bar{y}_j}{(\cos \phi)^d}\right)$$

- $d = |N(j)|$: number of CZ neighbors (varies per qubit)
- $\phi = 0$ recovers the original formula: $\hat{\theta}_j = \arccos(1 - 2\bar{y}_j)$

Estimation Formula: Interpretation

$$\hat{\theta}_j = \arccos\left(\frac{1 - 2\bar{y}_j}{(\cos \phi)^d}\right)$$

- The numerator $1 - 2\bar{y}_j \in [-1, 1]$ is the “signal” from the training data
- The denominator $(\cos \phi)^d$ **amplifies** the signal, compensating for the R_x -induced suppression
- Larger $d \Rightarrow$ smaller $(\cos \phi)^d \Rightarrow$ larger amplification
- For $\phi = \pi/4$: $\cos \phi = 1/\sqrt{2}$, so $(\cos \phi)^d = 2^{-d/2}$

This is a theoretically valid closed-form estimator — if the training data (x, y) were generated by the R_x -ISQNN circuit itself.

The question is: does this formula remain useful when y comes from an *external* data source (MNIST images)?

From Estimation to Inference: The Gap

The natural workflow:

- 1 Learn θ from training data (x, y) using the estimation formula
- 2 Use the learned θ to **infer** y for new test inputs x
- 3 Compare inferred y with test labels to verify correctness

The problem:

- Step 2 requires a circuit that accepts any x and produces y using the learned θ
- The only circuit whose distribution **matches** the estimation formula is the R_x -ISQNN ($n_1 m$ qubits)
- The m -qubit R_x -DQNN produces a **different distribution** (Reason 1)

Central question: Is there *any* efficient DQNN structure that can correctly infer using parameters learned from the R_x -ISQNN?

Overview: Two Reasons

- ① R_x **breaks the R_z -CZ commutation** — no classical rule can replicate ISQNN states
- ② **No low-qubit DQNN can replace ISQNN with R_x** — Catch-22

Reason 1: Original Equivalence (Without R_x)

Algebraic foundation: R_z and CZ **commute** — both diagonal in the Z basis.

$$[R_z \otimes I, CZ] = 0, \quad [I \otimes R_z, CZ] = 0$$

- $x = 1$ ($|0\rangle$): $R_z(\theta)|0\rangle = e^{-i\theta/2}|0\rangle$ (global phase). CZ eigenstate. **Stays** $|0\rangle$.
- $x = 0$ ($|+\rangle$): $R_z(\theta)|+\rangle = \cos \frac{\theta}{2}|+\rangle - i \sin \frac{\theta}{2}|-\rangle$. CZ entangles, but 50/50 in X .

Consequence: Product state structure $\Rightarrow$ each qubit's evolution is independently trackable. DQNN can replicate ISQNN's post-measurement state using classical prescriptions ($H+X$ for $x = 0$, $\text{measure}+\text{reset}$ for $x = 1$).

Reason 1 (cont.): R_x Breaks the Commutation Structure

R_x does **NOT** commute with R_z or CZ :

$$R_x(\phi) = \cos \frac{\phi}{2} I - i \sin \frac{\phi}{2} X \quad (\text{non-diagonal in } Z \text{ basis})$$

- $x = 1$: $|0\rangle \xrightarrow{R_x} |u\rangle$ — **no longer a CZ eigenstate**
- After $R_z(\theta)|u\rangle$ and CZ , the qubit is in a **complex entangled state**
- When inter-slice CZ acts, the qubit is **not** simply $|0\rangle$ or $|+\rangle$
- The evolution **cannot be factorized** into per-qubit tracks

For DQNN: Classical rules ($H + X$, reset, 50/50) assume the qubit is in a simple known state. With R_x , the actual state is complex and unknown to a classical prescription.

Reason 1 (cont.): DQNN's Irreconcilable Dilemma

In ISQNN with R_x , multiple $|u\rangle$ qubits connected by CZ form an **entanglement cluster** spanning slices. DQNN faces two options — both fail:

Reset to $ 0\rangle$ (current DQNN)	Don't reset
After measuring $x = 1$, reset removes $ u\rangle$ before subsequent CZ.	Keep qubit “alive” across blocks, accumulate all operations.
Result: Inter-slice $ u\rangle$ - $ u\rangle$ entanglement destroyed . DQNN sees isolated qubits.	Result: Output y depends on accumulated R_z +CZ history from $k - 1$ blocks, not just θ_j .

Dilemma: Reset = entanglement gone. No reset = history accumulated. Neither reproduces ISQNN.

Reason 2: No Low-Qubit-Count DQNN Exists for R_x

Without R_x : R_z and CZ commute $\Rightarrow$ product-state structure $\Rightarrow$ a classical rule based DQNN (m qubits) can reproduce the ISQNN ($n_1 m$ qubits) distribution exactly.

With R_x : Commutation broken $\Rightarrow$ qubit evolution is non-factorizable. The $|u\rangle$ - $|u\rangle$ entanglement clusters **cannot** be decomposed into a sequential measurement + reset circuit.

	Without R_x	With R_x
Algebraic structure	$[R_z, CZ] = 0$	$[R_x, CZ] \neq 0, [R_x, R_z] \neq 0$
Qubit tracking	1 qubit at a time	Must track all simultaneously
Low-qubit DQNN exists?	Yes (proven)	No (impossible)

The only correct circuit is ISQNN: $n_1 \times m = 100$ qubits, 2^{100} state vector $\Rightarrow$ **intractable**.

Reason 2 (cont.): The Catch-22

- MNIST requires $n_1 = 10$, $m = 10 \Rightarrow 100$ qubits
- ISQNN with R_x : 2^{100} state vector — computationally **impossible**
- Any DQNN with $m = 10$ qubits ($2^{10} = 1024$ dim): **provably wrong** with R_x

	ISQNN (R_x)	DQNN ($m = 10$, R_x)
State size	2^{100}	$2^{10} = 1024$
Correct distribution?	Yes	No (Reason 1)
Computationally feasible?	No	Yes

Catch-22: Correct circuit = intractable. Tractable circuit = incorrect. No middle ground exists.

- ① R_x breaks R_z -CZ commutation $\Rightarrow$ **no low-qubit DQNN** can replicate R_x -ISQNN
- ② ISQNN intractable (2^{100}); DQNN wrong with R_x — **Catch-22**

Restoring connectivity is not enough.

The R_x - R_z - CZ algebraic incompatibility breaks the product-state structure that made the original DQNN-ISQNN equivalence possible. Without that structure, no efficient classical-quantum hybrid DQNN can replace the full R_x -ISQNN — and the full ISQNN is computationally out of reach for 100 qubits.